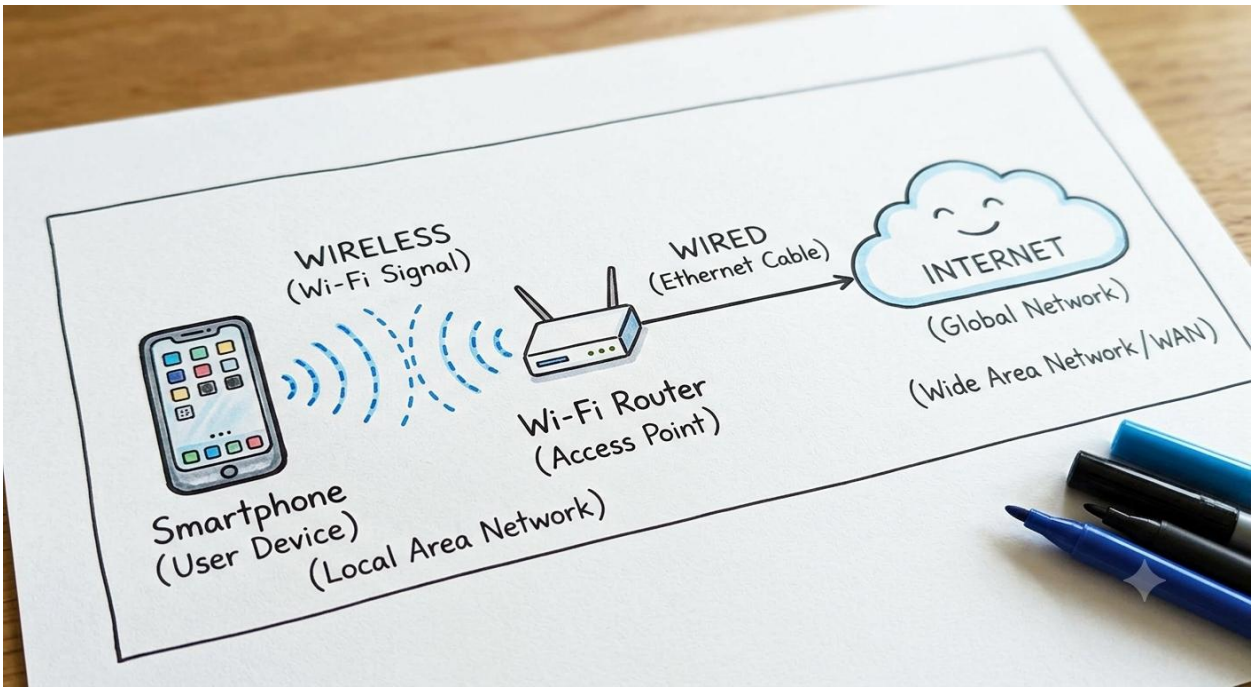
**1.List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.**

| Wi-Fi Standard | Frequency Band | Key Difference |
|---|---|---|
| 802.11n (Wi-Fi 4) | 2.4 GHz / 5 GHz | Speed up to 600 Mbps |
| 802.11ac (Wi-Fi 5) | 5 GHz | Faster, up to about 6.9 Gbps |
| 802.11ax (Wi-Fi 6) | 2.4 GHz / 5 GHz | Higher efficiency and speed, up to about 9.6 Gbps |

**2.Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network. generate an image**



**3.Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Hint: Consider both signal strength and security when choosing.**

Wi-Fi Network Scan

| SSID | Signal Strength | Security |
|---|---|---|
| Home_WiFi | Excellent (4/5) | WPA2 |
| JioFiber_5G | Good (3/5) | WPA2/WPA3 |
| TP-Link_5G | Fair (2/5) | WPA2 |

My Choice

I would choose JioFiber_5G because it has a good signal and uses WPA2/WPA3 security, which makes it a safer choice than an open or weakly secured network.

## 4.Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID hiding means making your Wi-Fi network name invisible when people scan for nearby Wi-Fi networks. The network is still there, but its name does not normally appear in the Wi-Fi list.

Advantage: It can make the network less noticeable to people nearby.

Disadvantage: It does not provide real security. Someone can still detect the network using Wi-Fi scanning tools, and devices that connect to it may reveal its name.